Quiz 01

```
CREATE OR REPLACE TYPE Empl_typ AS OBJECT
(
 empNo INT,
 empName VARCHAR2(50),
 salary FLOAT,
 designation VARCHAR2(50)
);
/

CREATE OR REPLACE TYPE Member_typ AS OBJECT
(
 team_member REF Empl_typ
);
/

CREATE TYPE Member_nt AS TABLE OF Member_typ;
/

CREATE TYPE Project_typ AS OBJECT
(
 projNo INT,
 pname VARCHAR2(50),
 members Member_nt,
```

```
 mgr REF Empl_typ
);
/

CREATE TABLE Employees OF Empl_typ(PRIMARY KEY(empNo));

CREATE TABLE Projects OF Project_typ(PRIMARY KEY (projNo), mgr REFERENCES
Employees)
NESTED TABLE members STORE AS member_tab;

INSERT INTO Employees VALUES (1, 'Sarath', 10000.0, 'Technician');
INSERT INTO Employees VALUES (2, 'Sampath', 12000.0, 'Sales Officer');
INSERT INTO Employees VALUES (3, 'Viraj', 18000.0, 'Software Developer');
INSERT INTO Employees VALUES (4, 'Tusith', 22000.0, 'Director');
INSERT INTO Employees VALUES (5, 'Nimali', 15000.0, 'Technician');

INSERT INTO Projects VALUES (10, 'Cabling Project',
member_nt(
Member_typ((select ref(e) from Employees e where e.empNo=1)),
Member_typ((select ref(e) from Employees e where e.empNo=5))),
null);

INSERT INTO Projects VALUES (12, 'Data Warehousing Proj',
member_nt(
Member_typ((select ref(e) from Employees e where e.empNo=2)),
```

```
Member_typ((select ref(e) from Employees e where e.empNo=3)),
Member_typ((select ref(e) from Employees e where e.empNo=4))),
(select ref(e) from Employees e where e.empNo=2)
);
```

Question 1

Print the project number, project name and project manager's salary. The results should be printed is descending order of salary values.

```
SELECT p.projNo, p.pname, m.team_member.salary
FROM Projects p
JOIN Employees m ON p.mgr = REF(m)
ORDER BY m.salary DESC;
```

Question 2

Write a member method (called Budget) that finds returns the total salary of all members in the project.

```
CREATE OR REPLACE TYPE Project_typ AS OBJECT
(
 projNo INT,
 pname VARCHAR2(50),
 members Member_nt,
 mgr REF Empl_typ,
```

```
 MEMBER FUNCTION Budget RETURN FLOAT
);
/


CREATE OR REPLACE TYPE BODY Project_typ AS
 MEMBER FUNCTION Budget RETURN FLOAT IS
  totalSalary FLOAT := 0.0;
 BEGIN
  FOR i IN 1..self.members.COUNT LOOP
   totalSalary := totalSalary + self.members(i).team_member.empNo.salary;
  END LOOP;

  RETURN totalSalary;
 END;
END;
/
```

Question 3

Use the method created above to find the project with the least budget greater than 20000. Print the projNo.

```
SELECT projNo
FROM Projects
WHERE Budget() > 20000
ORDER BY Budget()
FETCH FIRST 1 ROW ONLY;
```

Quiz 02

```
create type account_type_typ as object (
 code char(4),
 name varchar2(50),
 interest float,
 min_bal float
);
/

create type account_typ as object (
 account_no char(10),
 account_ty ref account_type_typ,
 balance float
);
/

create type account_tlb as table of account_typ;
/

create table account_types of account_type_typ (code primary key);

create type customer_typ as object (
 id char(10),
 name varchar2(50),
 accounts account_tlb
```

```
);
/

create table customers of customer_typ(id primary key)
 nested table accounts store as ntlb_accounts;

insert into account_types values ('SAVS', 'General Savings', 5.24, 100.0);
insert into account_types values ('CHEQ', 'General Checking', 0.0, 0);
insert into account_types values ('HAPS', 'Hapan Savings', 6.26, 20000.0);


insert into customers values (
 '781250401V',
 'Sampath Weerasinghe',
 account_tlb(
  account_typ('1122300004',
      (select ref(s) from account_types s where s.code = 'SAVS'), 5700.00),
  account_typ('2334453124',
      (select ref(s) from account_types s where s.code = 'CHEQ'), 2300.00)
 )
);

insert into customers values (
 customer_typ (
  '99334453V', 'Dulani Pieris',
  account_tlb(
```

```
  account_typ('3445663321',
          (select ref(s) from account_types s where s.code = 'HAPS'), 19045.06),
  account_typ('5000022123',
          (select ref(s) from account_types s where s.code = 'SAVS'), 55235.00)
 )
 )
);
```

Question 1

Deposit Rs. 1200/- to the account (Acc # 1122300004) of Mr. Sampath Weerasignhe (ID # 781250401V).

```
UPDATE TABLE(SELECT c.accounts FROM customers c WHERE c.id = '781250401V') a
SET a.balance = a.balance + 1200
WHERE a.account_no = '1122300004';
```

Question 2

Write a member function (called TotBal) that returns the total balance of a customer's accounts.

```
CREATE OR REPLACE TYPE customer_typ AS OBJECT (
 id CHAR(10),
 name VARCHAR2(50),
 accounts account_tlb,
 MEMBER FUNCTION TotBal RETURN FLOAT
```

```
);
CREATE OR REPLACE TYPE BODY customer_typ AS
 MEMBER FUNCTION TotBal RETURN FLOAT IS
  total_balance FLOAT := 0;
 BEGIN
  FOR i IN 1..self.accounts.COUNT LOOP
   total_balance := total_balance + self.accounts(i).balance;
  END LOOP;
  RETURN total_balance;
 END TotBal;
END;
/
```

Question 3

Write a query to print the names of customers and their total balances (using the function created in question 2)

```
SELECT c.name, c.accounts.TotBal() AS total_balance
FROM customers c;
```